

Dimensioning of Modular High Frequency Converter for drives

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Abstract— In order to meet the requirements of future electric drive systems, new converter topologies are investigated. Because light weight is one of the main requirements, the passive components (DC-capacitors, chokes, EMI-filters) have to be reduced drastically [1]. Additional important points are fault tolerance [2], EMI limits and the industrial scalability of the system. These requirements cannot be met with standard topologies [3][4]. The Modular High Frequency converter (MHF) is a new inverter topology for electrical drive systems. It is well adapted to those future requirements, especially ultra-light weight, small volume, low losses, fault tolerance and an excellent dynamic behavior [5][6][7]. In order to optimize the drive system, using a MHF converter, a good adaption of the MHF converter to the parameters of the PM-Machine and vice versa is necessary. This paper deals with the design parameters and presents the dimensioning of the converter for a typical permanent magnet machine. First measurements of a laboratory prototype are depicted and the newest version of the power circuit is shown.

Keywords— *Modular High Frequency Converter; MHF, multi-level converter; inverter for drive Systems; high switching frequency; low voltage semiconductor*

I. BASIC FEATURES OF THE MODULAR HIGH FREQUENCY CONVERTER TOPOLOGY

The MHF converter consists of a freely chosen number of identical submodules (SM), which enable independent control of the DC- and AC-Side values (Fig. 1). The AC-Sides of the submodules control the currents in the windings of the machine, separately. The DC-Sides of all SM and an extremely small DC filter choke form the DC input terminals (P, N) of the converter. There is no need for DC-Bus capacitors avoiding high surge currents in case of failures and resonance currents in the DC-Bus Network [3]. The new DC configuration enables fast and direct control of the battery current by controlling the multilevel DC-Input voltage $\sum V_{CM}$. The new modular topology allows small and precisely controlled internal voltages per submodule ($V_C < 200V$) enabling the use of Low Voltage semiconductors. Low voltage MOSFETs feature very low losses and an extended maximum junction temperature. In addition, the high switching frequencies of these semiconductors enable the use of small ceramic chip capacitors (instead of foil or electrolytic caps). The Low

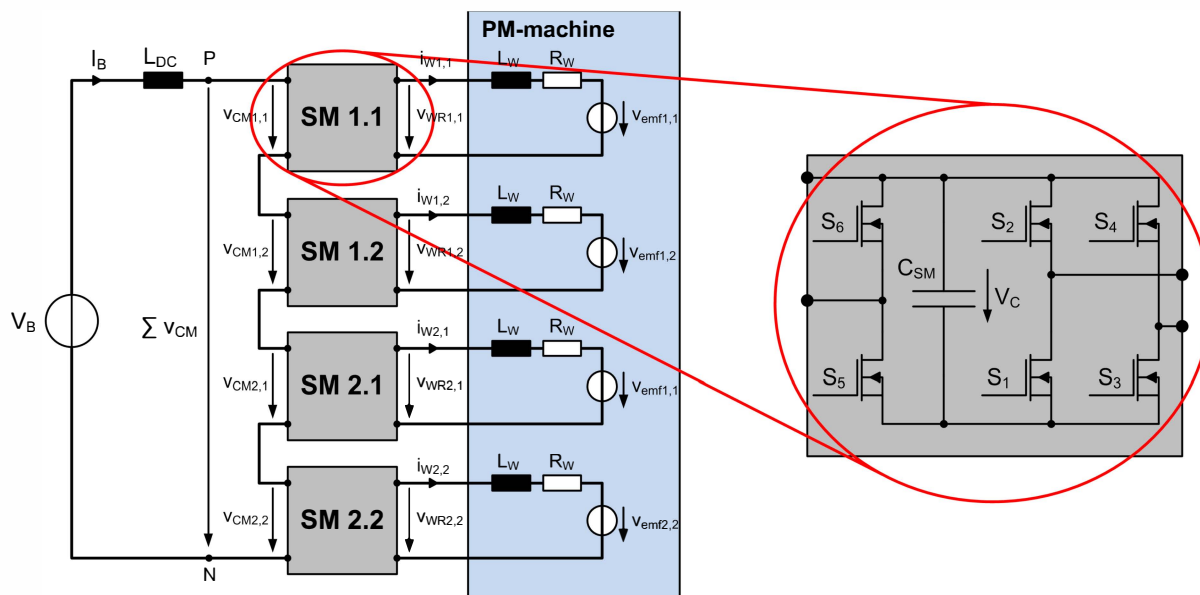


Fig. 1. Modular High Frequency Converter (MHF); example for $n_{SM} = 4$ submodules

The research is funded by the German Federal Ministry of Education and Research

Voltage semiconductors and the absence of heavy DC-Busbars and DC-Bus capacitors create suitable conditions for monolithic integration and motor integration. The efficiency of the MHF converter in comparison to an IGBT 2-level converter has been investigated in [4]. Typically, the MHF offers a superior efficiency at partial load, which is an important operating range in most vehicle applications. At full load, however, no significant difference of efficiency is present. With regard to the future development of low FETs (Silicon or GaN) a further increase of efficiency can be foreseen. An estimation of the switching losses in a MHF submodule is given in Chapter V.

II. DIMENSIONING OF MHF CONVERTER FOR A PERMANENT MAGNET MACHINE

In a first step, for a given machine and a given DC-voltage, the number of submodules can be chosen. In order to simplify the explanation, a conventional 3-phase star connected machine (originally designed for 2-level VSC) is taken as the starting point (Table I). For such a conventional drive system the minimum DC-Bus voltage V_{DC} for unrestricted operation is an important design parameter. For the new converter system, the coils of the machine must be adapted (number of turns N) to match the internal submodule voltage V_C . Therefore, the voltage ratio K , according to (1) can be computed. It is defined by the ratio of the minimum DC-Bus voltage (V_{DC}) and the internal (stabilized) DC-Voltages of the submodules (V_C). In this example, the voltage ratio K is about 2.

TABLE I. INITIAL VALUES: PERMANENT MAGNET MACHINE (3 PHASE-STAR-CONNECTION)

Parameter	
Power, nominal - P_{nom}	33kW
Speed, nominal - n_{nom}	7400rpm
Winding current, nominal - $I_{w,nom}$	100A
Excitation flux - Ψ_{PM}	0.0228Wb
Winding resistance - R_w	20.6mΩ
Winding Inductance - L_w	0.177mH
Poles	12
Number of phases - n_{phase}	3
DC-Bus Voltage - V_{DC}	310V

TABLE II. INITIAL VALUES: PERMANENT MAGNET MACHINE (3 PHASE-STAR-CONNECTION)

Parameter	6 Submodules	9 Submodules ($N_{new} = 1.5 \cdot N$)
Number of Submodules n_{SM}	6 (2 per phase)	9 (3 per phase)
Submodule Power P_{nom} / P_{max}	5.5kW / 6.6kW	3.66kW / 4.4kW
Submodule Voltage V_C	100V	100V

Output current, nominal $I_{w,nom} / I_{w,max}$	102A / 153A	68A / 104A
Excitation flux Ψ_{PM} (per phase)	0.0115Wb	0.0115Wb
Winding resistance R_w	10.25mΩ	10.25mΩ
Winding Inductance L_w	0.088mH	0.132mH
Maximum battery Voltage $V_{B,max}$	300V	450V

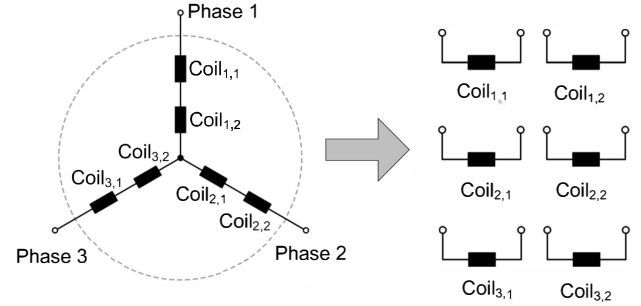


Fig. 2. Voltage level adapting by splitting the coils of a conventional machine (example for $n_{SM} = 6$)

This means, that in comparison to the conventional machine, the voltage level of the machine coils must be halved. The voltage level can be adapted by splitting the coils into several smaller ones and/or adapting the winding number of the coils (Fig. 2). Table II shows the parameters of two different MHF converter designs with six (splitted coils) or nine (splitted coils and adapted winding number N) submodules. In order to fully exploit the advantages of the MHF concept, a higher number ($n_{SM} \geq 4$) of submodules is favorable in most cases. The main advantages are a higher number of DC voltage levels and a higher effective switching frequency on the DC-Side. The ability for redundant operation and fault ride through in case of defective submodules [5] is improved, too. Fig. 3 shows the current waveform of the AC-Side of one submodule at $P = P_{max}$ and nominal speed (switching frequency $f_p = 150kHz$).

$$K = \frac{V_{DC}}{V_C \cdot \sqrt{3}} = \frac{310V}{100V \cdot \sqrt{3}} \approx 2 \quad (1)$$

III. DIMENSIONING OF THE SUBMODULES

A. Minimization of the conduction losses

For FET based converters, increasing the number of paralleled FETs per submodule reduces the conduction losses essentially. On the contrary - for IGBT converters - there is only marginal improvement when the silicon area is increased. A further loss reduction can be achieved by shifting FET-Chips from the high to the low side (asymmetric chip areas in the phase leg). In this case, by restricting the zero voltage vector switching state to the low side switches, will result in lower conduction losses. This specific measure, however, would not be advantageous for

IGBT based converters. Furthermore, the diode losses of a FET-based system can be eliminated by the gate controlled n-channels of the FETs for shunting the body diodes. This advantage can also be realized - with reasonable expense - for low voltage FETs, only.

B. Dimensioning of the internal submodule capacitor C_{SM}

The ability to operate without DC-Bus capacitors and with extremely small capacitance in the submodules (C_{SM}) is a big advantage of the MHF concept. For the dimensioning of the internal submodule capacitance C_{SM} , the instantaneous power flow in a submodule must be considered. As stated before, the AC- and DC-Side of the MHF converter are controlled independently. The AC-Power ($i_W \cdot v_{emf}$) of a submodule depends on the desired winding current i_W and the actual emf voltage (speed) of the machine. The control unit of the MHF converter adjusts the DC-Power ($I_B \cdot a_{cx,y} \cdot V_{Cx,y}$) of each submodule, in order to fulfill two main conditions:

- The DC-Power of each submodule equals the corresponding AC-Power at each instant of time.
- The sum of the DC-Powers of all submodules is constant at each instant of time (no ripple)

The remaining small differences between the AC-Power and the DC-Power are absorbed by the internal submodule capacitance C_{SM} . The most significant differences appear in the small time span with reverse power flow (φ_{WR} in Fig. 3). In motor operation this is the time span, where instantaneous negative powers occur (in generator operation: instantaneous positive power). The time span is a function of the power factor $\cos(\varphi_{WR})$ of the converter, which can be chosen within certain limits. Typically, the worst case operating point with respect to this problem occurs at nominal speed and maximum power. Fig. 5 shows the instantaneous power flow in this operating point of the MHF converter with six submodules (Table. II). The hatched areas in the middle subplot correspond to the energy that must be absorbed by the capacitance of each submodule. In good approximation the required capacitance can be computed using (2)-(4) (This approximation holds for $|\varphi_{WR}| < 36^\circ$). It depends on the phase angle of the converter φ_{WR} , the allowed internal voltage ripple ΔV_C , the output current I_W and voltage V_W magnitude and the fundamental output frequency f_1 . The results of simulation (fig. 5) are compared to the results from (2)-(4) in Table III. As can be seen, very small capacitances are sufficient. By using 100V SMD ceramic chip capacitors (X7R) a capacitance of $C = 300\mu F$ corresponds to a total weight of approximately 50 grams.

$$W^+ = W^- = \Delta W \approx \frac{\hat{I}_W \cdot \hat{U}_W}{2\pi \cdot f_1} \cdot \frac{\varphi_{WR}^3}{6} \quad (2)$$

$$A = \frac{\Delta V}{V_{C,nom}} \quad (3)$$

$$C_{SM} = \frac{2 \cdot \varphi_{WR}^3 \cdot \hat{I}_W \cdot \hat{U}_W}{V_{C,nom}^2 \cdot 12A \cdot 2\pi \cdot f_1} \quad (4)$$

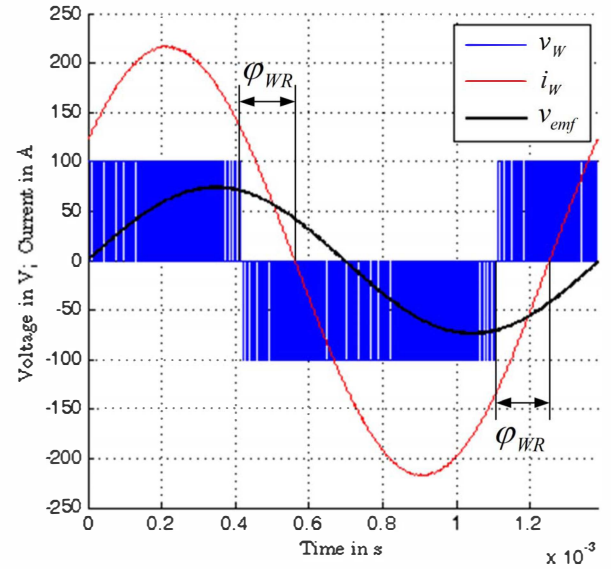


Fig. 3. AC-Side of one submodule at $P = 6.5kW$ and $n = 7200$ rpm

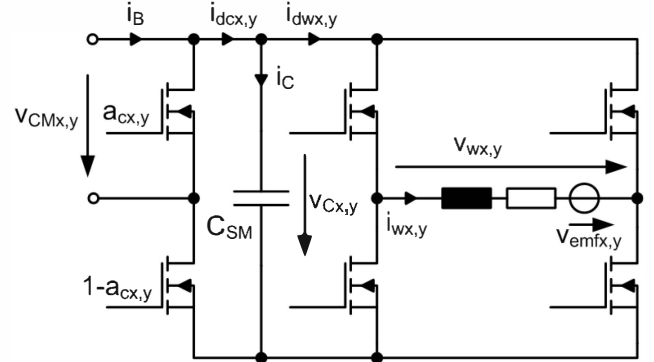


Fig. 4. Definition of submodule signals

TABLE III. REQUIRED SUBMODULE CAPACITANCE C_{SM} : SIMULATION VS. CALCULATION

$A = 0.05, P = 6.5kW, n = 7200rpm, \varphi_{WR}=35^\circ$		
	Calculated C_{SM}	Simulated C_{SM}
MHF: 6 submodules	297 μF	308 μF
MHF: 9 submodules	217 μF	215 μF

IV. DIMENSIONING OF THE BATTERY VOLTAGE AND THE DC FILTER CHOKE

The minimum operating DC-Voltage in a system using a MHF converter is limited by the maximum allowable battery current (In principle not by the MHF converter). The maximum battery voltage $V_{B,max}$ of the drive system depends on the number of submodules n_{SM} and the chosen submodule voltage V_C (5). The operation and functionality of the MHF converter is independent of the actual battery voltage in a wide range.

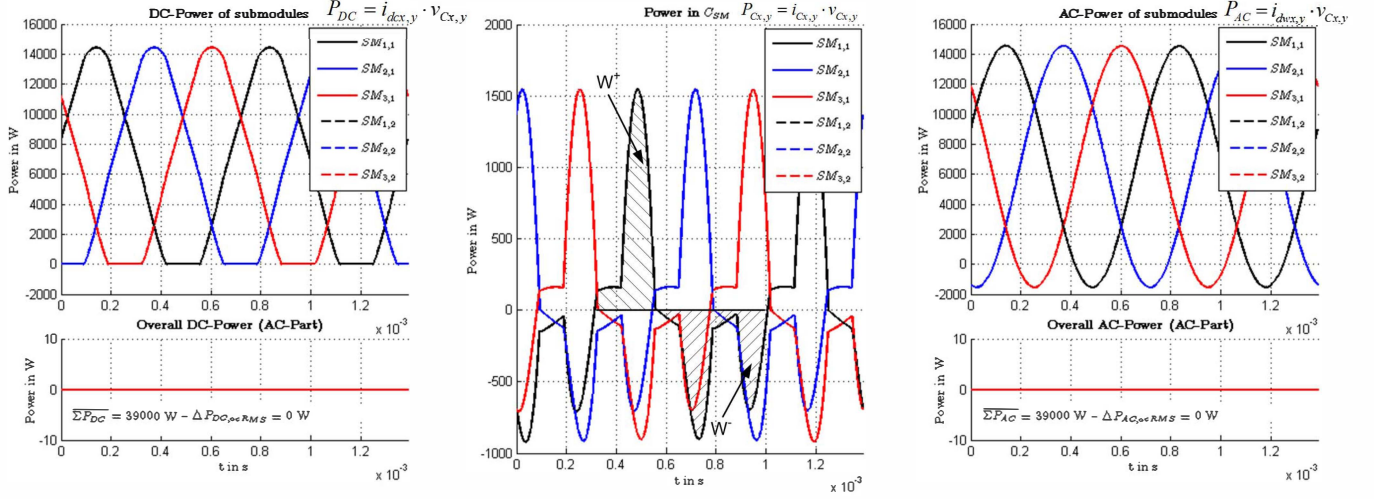


Fig. 5. MHF converter with 6 submodules: instantaneous power flow of the DC-Side (left), capacitor (middle), AC-Side (right)

$$V_{B,max} = \frac{n_{SM} \cdot V_{C,nom}}{2} \quad (5)$$

Assuming a constant instantaneous overall AC-Power (Sum of all AC-Powers), the overall instantaneous DC-Power (Sum of all DC-Powers) of the MHF converter is constant (Fig. 5). Therefore, the ripple components of the battery current are restricted to multiples of the PWM frequency. An additional phase shifting of the PWM signals of the submodules leads to a very high switching frequency $f_{p,DC}$ and a multilevel DC-Voltage $\sum V_{CM}$ on the DC-Side of the converter (6).

$$f_{p,DC} = f_{p,SM} \cdot n_{SM} \quad (6)$$

In comparison to a conventional converter that uses two voltage levels ($n=2$) a higher number of voltage levels (n) leads to a significant reduction of the required DC inductance (L_n) [7]. Assuming the same switching frequency f_p of the semiconductors, the required DC filter inductance decreases with $1/n^2$ (Fig. 6: red curve). A high number of voltage levels enable the use of lower voltage semiconductors and therefore higher switching frequencies f_p . If the dv/dt of the semiconductors is kept constant, the required inductance decreases even more (Fig. 6: blue curve), whereas the EMI disturbances are on kept on a comparable level. Thus, only an extremely small DC filter choke L_{DC} at the battery side is needed in order to limit the AC ripple current component.

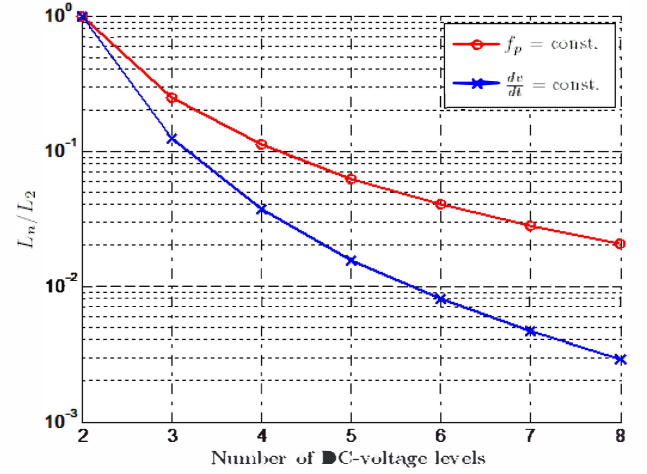


Fig. 6. Required inductance of the DC filter choke (referred to 2-level Chopper)

V. MEASUREMENT RESULTS AND SUBMODULE POWER CIRCUIT

A. Steady state operation

First measurements of a small laboratory version of a MHF converter with four submodules ($V_C = 60V$) were done. Passive loads (ohmic-inductive) at the AC-Sides of the submodules were used for this test. The measurement was done by using a DC filter choke of $L_{DC} = 5\mu H$, only. No additional filters were used. The battery voltage in this example was $V_B = 120V$. The switching frequency of each submodule was $f_p = 100kHz$. Fig. 7 shows three of four AC currents and the DC current I_B in steady state operation at $P = 5kW$ and $f_l = 200Hz$ (display limited by number of oscilloscope channels). The mean value of battery current (10A/div) is $I_B = 46A$, the rms value of the ripple current component is 1.9A.

The Fourier analysis of the battery current is shown in Fig 8. According to the expectations there are just small interfering frequencies (1.2A@100kHz, 0.6A@200kHz) at multiples of

the switching frequency f_p . Multiples of the fundamental frequency f_1 are not expected under ideal conditions. They can occur, however, due to small fluctuations of the overall instantaneous AC-Power. This point leads to high accuracy requirements of the digital control system.

B. PCB-based submodule power circuit

1) Layout

There are several options to realize the power circuit of the MHF converter. Besides a DCB (Direct Copper Bond) design, it is possible to use a multi-layer PCB (Printed Circuit Board). The present trend for leadless MOSFET packages (DirectFET®, etc.) without or minimized bonding wires enables PCB layouts with very small parasitic inductances. The MOSFET drivers can be implemented on the same PC-board, in order to achieve minimum switching times (by minimizing the inductance of the gate circuit). Fig. 9 depicts a realized design of a PCB-based submodule power circuit. The half bridges of the module are asymmetric with twice the number of FETs on the low-side compared to the high side. Table IV shows the parameters of the prototype.

2) Switching losses

Fig. 10 shows a switching procedure of one MOSFET half bridge. The low-side FET turns on, whereas the high-side FET turns off. The overshoot voltage of the Drain-Source voltage of the high-side MOSFET $V_{DS,high}$ is below 5% of the

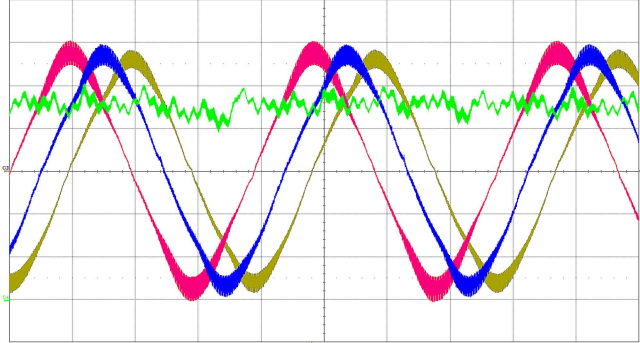


Fig. 7. MHF Converter in steady state operation (1ms/div): Yellow, Blue, Red: AC-Currents (50A/div); Green: Battery Current I_B (10A/div)

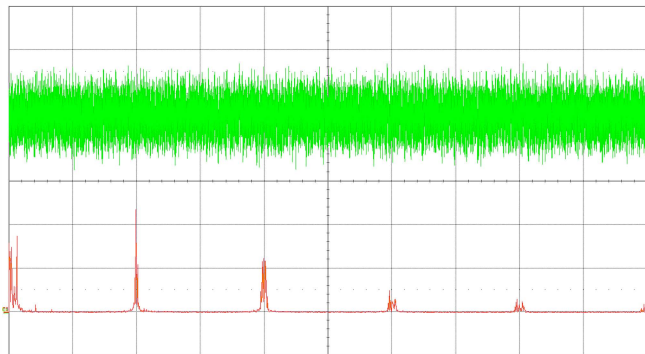


Fig. 8. Battery Current and its amplitude spectrum: Green: Battery Current I_B (20ms/div, 10A/div); Orange: Amplitude Spectrum (10kHz/div, mean value suppressed, 0.5A/div)

submodule voltage V_C . The implemented interlock time t_i between the low- and the high-side switch was chosen to be very small (approx. 3.3ns between the control signals). In order to understand the switching process and switching losses, Fig. 11 shows the turn-off process of the high side FET with increased interlock time (approx. 30ns). First, the drain current of MOSFET commutated from the n-channel to the parallel parasitic chip capacitor in an approximately linear manner. The time t_f is very short and barely visible in the oscillogram (assumption: $t_f = 3ns$). The switching losses during the time t_f can be calculated by (7)-(10)

$$i_D(t) = \hat{I}_W \cdot \left(1 - \frac{t}{t_f}\right) \quad (7)$$

$$u_{DS}(t) = \frac{\hat{I}_W \cdot t_f}{2 \cdot C_{oss,HB}} \cdot \left(\frac{t}{t_f}\right)^2 \quad (8)$$

$$P_{SW} = f_p \cdot \int_0^{t_f} i_D(t) \cdot u_{DS}(t) \cdot dt \quad (9)$$

$$P_{SW} = f_p \cdot \frac{\hat{I}_W^2 \cdot t_f^2}{24 \cdot C_{oss,HB}} \quad (10)$$

If using the output charge Q_{oss} that is given by the datasheet, the chip capacitance of a half bridge $C_{oss,HB}$ will be about 23nF. Strictly spoken, an energy related chip capacitance has to be taken. The switching loss ($f_p = 100kHz$) during t_f is about 3mW and therefore almost neglectable. After the time interval t_f the load current has completely commutated into the chip capacities of the FETs and the voltage across the switch $V_{DS,high}$ rises linear. At the end of the interlock time (Fig. 11: right cursor) the low-side FET turns on the dv/dt of the drain source-voltage $V_{DS,high}$ increases immediately. The low-side FET dissipates the energy which is stored in the parasitic chip capacitors of all FETs. These losses cannot be completely avoided. A good linear approximation of the effective parasitic chip capacitance of a half bridge $C_{oss,HB}$ is also achievable by measuring the dv/dt of the drain source voltage $V_{DS,high}$ during the interlock time (11).

$$C_{oss,HB} = I_W \frac{\Delta t}{\Delta V_{DS}} = 24nF \quad (11)$$

The switching losses of one half bridge P_{SWC} caused by the dissipation of the capacitor charge can be calculated by (12).

$$P_{SWC} = f_p \cdot V_C^2 \cdot C_{oss,HB} \quad (12)$$

According to (12), switching with $f_p = 100kHz$ and a submodule voltage $V_C = 80V$ results in switching losses of about 16W per asymmetrical half bridge.

TABLE IV. PARAMETER OF SUBMODULE PROTOTYPE (2ND VERSION)

Parameter	
Nominal Submodule Voltage $V_{C,nom}$	80V
Submodule Capacitance C_{SM}	1.2mF
Maximum blocking Voltage $V_{DS,max}$	120V
$R_{DS,on}$ High-Side FETs	0.96m Ω
$R_{DS,on}$ Low-Side FETs	0.48m Ω
Maximum gate current ON $I_{G,on,max}$	2.5A per FET
Maximum gate current OFF $I_{G,off,max}$	3.5A per FET
Dimension	142mm x 82mm

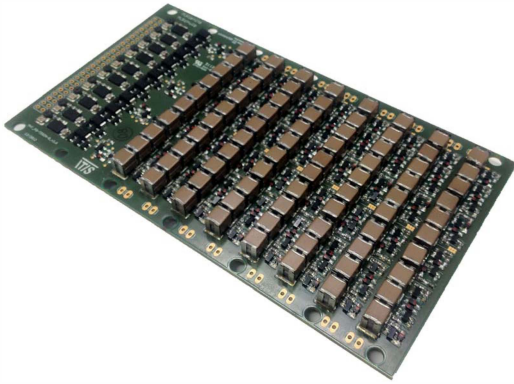


Fig. 9. Top-view of the realized submodule

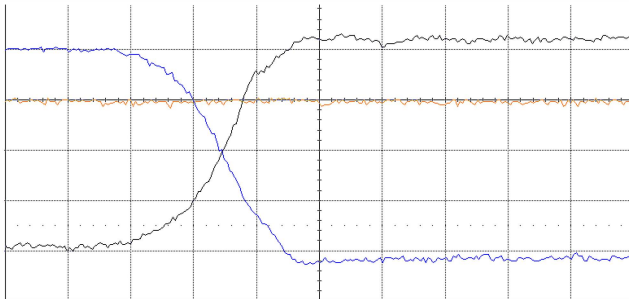


Fig. 10. Switching procedure of one half bridge ($t_i = 3.3ns$): $I_W = 60A$, $V_C = 80V$ (5ns/div): Yellow: I_W (20A/div); Blue: $V_{DS,low}$ (20V/div); Grey: $V_{DS,high}$ (20V/div)

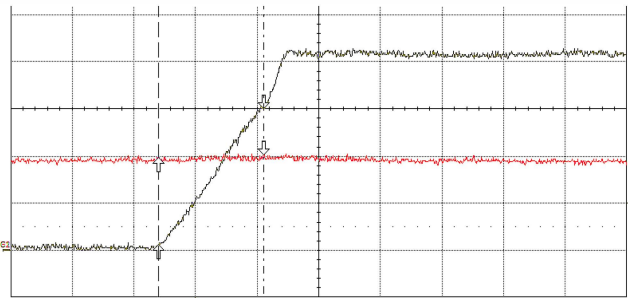


Fig. 11. Switching procedure with increased interlock time ($t_i = 30ns$): $I_W = 40A$, $V_C = 80V$, (20ns/div): Red: I_W (20A/div); Gray: $V_{DS,high}$ (20V/div); Blue: $V_{DS,low}$ (20V/div)

VI. CONCLUSION

The MHF converter offers superior technical characteristics. In comparison to conventional 2-level converter systems the weight and volume of the passive components are drastically reduced which results in ultra-light weight and a small volume converter system. The paper explains, how to dimension the required submodule capacitance C_{SM} and the DC filter choke. Additionally, the MHF features low losses and “fault ride through” capability during electronic defects. The modular design is industrially scalable and suitable for direct integration into electric machines. The Modular High Frequency converter is mainly intended for synchronous machines with permanent magnets.

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